APPENDIX 2

USER INFORMATION FOR THE ICAO-ACR COMPUTER PROGRAM

1.0 General Information. The ICAO-ACR computer program is maintained by the US Federal Aviation Administration (FAA), William J. Hughes Technical Center (WJHTC). The program implements the ACR computational procedures for rigid and flexible pavements. ICAO-ACR incorporates LEAF (Layered Elastic Analysis – FAA), a computer program that computes the structural responses of a layered pavement system according to Burmister's theory (layered elastic model). ICAO-ACR is distributed in compiled form as a Visual Basic.NET dynamic-link library (DLL). Program files may be downloaded from the FAA WJHTC web site:

http://www.airporttech.tc.faa.gov/Products/Airport-Pavement-Software-Programs

The following files are available for download from the above web site:

a) ICAO-ACR is an executable (stand-alone) computer program that executes the DLL *ACRClassLib.dll*, and returns standard ACR values.
b) *ACRClassLib.dll* is a Visual Basic.NET DLL that can be linked directly to other programmes that either compute ACR directly, or that use the ACR computation to evaluate PCR. *ACRClassLib.dll* is not a stand-alone computer program. Rather, it is intended to be run from within a separate calling program such as ICAO-ACR. Information on linking the library to a calling program is given below.

ICAO-ACR is an open-source program. The source codes for ICAO-ACR, ACRClassLib.dll and LEAF may be obtained from:

Federal Aviation Administration
William H. Hughes Technical Center
Airport Technology R&D Branch., ANG-E26
Atlantic City International Airport, NJ 08405
USA

2.0 Dynamic-Link Library (DLL) Technical Information.

The DLL *ACRClassLib.dll* was compiled using Microsoft Visual Basic 2013 in the Microsoft Visual Studio programming environment. Its target framework is Microsoft .NET Framework 4.5.

2.1 Input Data. The *ACRClassLib.dll* class library accepts the following data inputs:

2.1.1 Aircraft gross weight (in tonnes or pounds).
2.1.2 Percent of aircraft gross weight acting on the main gear, expressed as a decimal value.
2.1.3 Number of wheels in the aircraft gear to be analysed.
2.1.4 Tire pressure (in MPa or pounds per square inch).

2.1.5 Horizontal coordinates (x, y) of each wheel (in mm or inches).
2.1.6 For each wheel, a value 0 or 1, indicating whether the wheel is within the limits of the evaluation point grid. (The value 1 indicates that it is included.)
2.1.7 Pavement type. This value can only be "Flexible" or "Rigid."
2.1.8 System of Units (Metric or US).

2.2 Output data. The ACRClassLib.dll class library returns the following data outputs, given the above inputs:

2.2.1 ACR thickness *t* corresponding to four standard subgrade categories (in inches).
2.2.2 ACR number corresponding to four standard subgrade categories.

2.3 Procedure for linking to DLL. *ACRClassLib.dll* is a .NET DLL. For projects compiled in the Microsoft Visual Studio.NET programming environment, the procedure for linking to the DLL is as follows:

2.3.1 In the project properties, add *ACRClassLib.dll* to References.
2.3.2 Declare all variables that will be passed between the calling program and *ACRClassLib.dll*. The following input variables are declared as Single type: aircraft gross weight, percent gross weight, tire pressure, x-coordinate (array), y-coordinate (array). The following input variables are declared as Integer type: number of wheels, wheel selection variable (array) (see 2.1.6).

Certain variables have special definitions. The pavement type is specified as an enumerate variable type:

```
Public Enum PavementType
    Flexible = 1
    Rigid = 2
End Enum
```

ACR data are stored in a Visual Basic data structure ACRdata:

```
Public Structure ACRdata
    Dim libACR() As Single
    Dim libACRthick() As Single
    Dim libSubCat() As String
    Dim libSubCatMPa() As String
End Structure
```

The four elements in data structure ACRdata are:

1. `ACRdata.libACR()` stores ACR numerical values following execution.
2. `ACRdata.libACRthick()` stores ACR thickness values following execution.
3. `ACRdata.libSubCat()` stores subgrade category letter designations.
4. `ACRdata.libSubCatMPa()` stores subgrade category standard modulus values in MPa.

Each of the elements 1-4 above is an array of length 5, of declared data type as indicated above. Within each array, the first element in the array `ACRData.array(0)` is not used, while the last four elements `ACRData.array(1)` through `ACRData.array(4)` correspond to standard subgrade categories "D" through "A" respectively. For reference, the following snippets of Visual Basic code are examples of function signatures used in the DLL. Executing the function `CalculateACR` from the calling program returns ACR values in the array `ACRdata.libACR()`. The first function signature applies for the majority of gear types where all wheels in the main gear have equal tire pressure and load. The second function signature is used specifically to compute the flexible ACR for certain gear configurations (e.g., the Airbus A340 series) where the center landing gear has a different tire pressure/wheel load combination than the wing landing gear.

```
Public Overloads Function CalculateACR(ByVal PavementType As clsACR.PavementType, _
       ByVal gross_weight As Single, _
       ByVal percent_gw As Single, _
       ByVal wheels_number As Integer, _
       ByVal tire_pressure As Single, _
       ByVal CoordX() As Single, _
       ByVal CoordY() As Single, _
       ByVal SW() As Integer, _
       ByVal Metric As Boolean) As ACRdata

Public Overloads Function CalculateACR(ByVal PavementType As clsACR.PavementType, _
       ByVal gross_weight As Single, _
       ByVal percent_gw As Single, _
       ByVal wheels_number As Integer, _
       ByVal tire_pressure As Single, _
       ByVal CoordX() As Single, _
       ByVal CoordY() As Single, _
       ByVal percent_gw2 As Single, _
       ByVal wheels_number2 As Integer, _
       ByVal tire_pressure2 As Single, _
       ByVal CoordX2() As Single, _
       ByVal CoordY2() As Single, _
       ByVal Metric As Boolean) As ACRdata 'ACR for two gears
```

The following is sample Visual Basic code for declaring variables in the calling program:

```
Dim ACRData As ACRClassLib.clsACR.ACRdata
Dim PavementType As ACRClassLib.clsACR.PavementType
Dim Gross_Wt, Percent_GW, Tire_Pressure As Single
Dim No_Wheels As Integer
Dim X1(), Y1() As Single
Dim SW() As Integer
Dim Metric As Boolean
```

2.3.3 Assign numerical values to declared input variables. The Boolean variable `Metric` is `True` for metric units, `False` for US units. Figure D-1 explains how to use variable `SW()`, which tells the program whether to include a given wheel in the limits of the evaluation point grid (see Appendix C, 1.1.3.7 (d)). In ICAO-ACR, a value of 1 assigned to SW means the wheel is included in the strain evaluation point grid area; any value other than 1 is treated as 0. Note that *ACRClassLib.dll* does not determine the correct number of wheels to include in the strain evaluation point grid. This determination should be made by the calling program with reference to the guidance in Appendix C, 1.1.3.7 (d). Also note that variable `SW` only controls which sub-group of wheels in the main gear assembly defines the strain evaluation point grid, not the number of wheels used to determine ACR. *ACRClassLib.dll* determines the ACR value for all wheels passed to it (`No_Wheels`). (In Figure D-1, if all wheels 1-8 were assigned `SW` = 1, the difference in computed ACR values would be insignificant. However, the computation would take much longer.)

2.3.4 Call Function `CalculateACR`. The following snippet of Visual Basic code calls the function `CalculateACR`.

```
 Dim RunACR As ACRClassLib.clsACR
        RunACR = New ACRClassLib.clsACR()

        ACRData = RunACR.CalculateACR(PavementType, Gross_Wt, Percent_GW, No_Wheels,
Tire_Pressure, X1, Y1, SW, Metric)
```

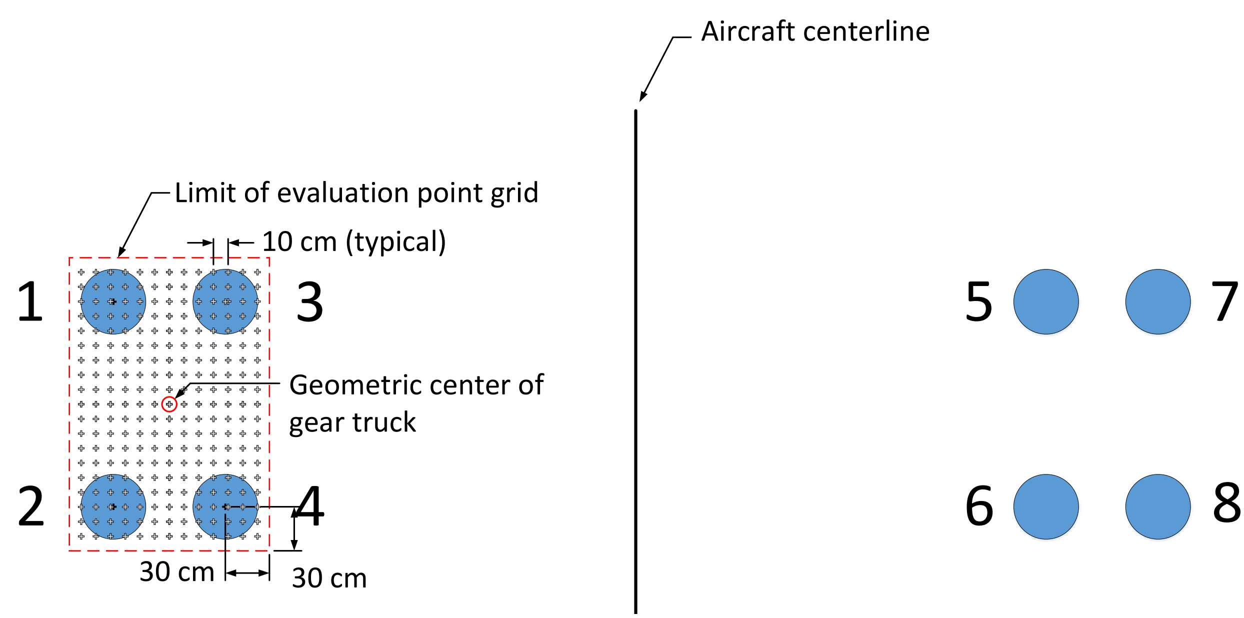

Figure D-1. Explanation of Variable SW for 2D Aircraft Gear.

$$\mathtt{SW} = \begin{cases} 1(wheels\ 1-4) \\ 0(wheels\ 5-8) \end{cases}$$

2.4 Example of Code. The following Visual Basic source code provides an example of how to call *ACRClassLib.dll* to calculate rigid ACR values for a generic dual-tandem (2D) aircraft gear. Note that for rigid ACR only the four wheels of one truck are used.

Also note that the index values for the tire geometric coordinates `X()` and `Y()` are 1 through 4.

```
'Example 1 in VB.NET
'Calculation of ACR values for 2D-400 on rigid pavements
'US units
'File ACRClassLib.dll needs to be included in Project References

Dim PavementType As ACRClassLib.clsACR.PavementType
Dim gross_weight As Single
Dim percent_gw As Single
Dim wheels_number As Integer
Dim tire_pressure As Single
Dim X1(4), Y1(4) As Single
Dim metric As Boolean

Dim ACRdata As ACRClassLib.clsACR.ACRdata

gross_weight = 400000 'lbs
percent_gw = 0.475
wheels_number = 4
tire_pressure = 200 'psi

X1(1) = -15 : Y1(1) = 0.0 'inches
X1(2) = 15 : Y1(2) = 0.0
X1(3) = -15 : Y1(3) = 55.0
X1(4) = 15 : Y1(4) = 55.0

PavementType = ACRClassLib.clsACR.PavementType.Rigid
metric = False

Dim RunACR As ACRClassLib.clsACR
RunACR = New ACRClassLib.clsACR()

ACRdata = RunACR.CalculateACR(PavementType, gross_weight,
          percent_gw, wheels_number, tire_pressure, X1, Y1, metric)

RunACR = Nothing

'Output ACR results
'ACRdata.libACR(1) = 894.672058
'ACRdata.libACR(2) = 817.8752
'ACRdata.libACR(3) = 744.0264
'ACRdata.libACR(4) = 641.6398
```

3.0 Program ICAO-ACR.

Program ICAO-ACR functions as a stand-alone program that computes flexible and rigid ACR values for arbitrary aircraft gear configurations, using the *ACRClassLib.dll* DLL. For convenience, the program includes a library of aircraft types commonly in use. For library aircraft, program ICAO-ACR automatically selects the correct number of wheels for ACR evaluation, i.e., all wheels in the main landing assembly for flexible ACR and all wheels in the most demanding single truck for rigid ACR.